Motion-Adaptive Internal Force Allocation for Dual-Arm Cooperative Manipulation // Or // Task-on-Demand Grasp Force Optimization for Dual-Arm Object Transport

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***Abstract*—**

- **Cooperative dual-arm transport needs enough squeeze force to prevent slip, but constant or over-conservative squeeze force wastes actuation effort and risks damaging fragile objects.**
- **We decompose the 12D dual-arm contact wrench into internal (squeeze) and external (task) components via a nullspace projection of the grasp matrix.**
- **We formulate a real-time 2-variable QP that allocates per-arm normal forces, minimizing squeeze effort and balancing the load between arms subject to friction-derived bounds.**
- **We couple the minimal grasp-force bound to the motion state of the manipulation task ("task-on-demand"), so the safety margin grows only while the object is in transit and relaxes at rest.**
- **We extend the same QP, at no extra decision-variable cost, with a joint-torque-aware penalty that biases squeeze-force allocation toward the arm with more actuation margin.**
- **We report internal force tracking error, friction-margin utilization, and squeeze effort with vs. without the motion-adaptive term, and with vs. without the joint-torque-aware penalty.**

***Index Terms*—ToDo [1]**

INTRODUCTION

Motivation

- Dual-arm cooperative transport needs sufficient normal (squeeze) force to prevent slip under load/inertial disturbance.
- Excess squeeze force is wasted actuation effort and risks damaging or destabilizing fragile/compliant objects.
- Human grip-force control scales with load/motion state rather than staying constant [to cite]: Westling & Johansson 1984 (grip force / load force coupling with friction-dependent safety margin).

Positioning

- Cooperative-manipulation force-distribution literature optimizes internal force statically or per-instant, without a motion-state-dependent target.
- Motion/slip-adaptive grip-force literature is mostly single-gripper and slip-reactive (triggered after slip is detected/predicted), not posed as a dual-arm allocation problem.
- Force-distribution literature rarely couples the allocation decision back to each arm's actuation margin (joint torque headroom); it is typically treated at the end-effector/wrench level only.

Gap

- No existing framework couples: (a) nullspace-based internal/external wrench decomposition, (b) real-time QP allocation of squeeze force between two arms, and (c) a motion-derived force demand that is proactive (scheduled/estimated from the task's motion state) rather than reactive (slip-triggered).
- Joint-space actuation capacity (torque headroom, proximity to singularity/thermal limits) is not typically fed back into a real-time end-effector-level force allocator, despite the wrench-to-torque map being affine and cheap to exploit.

Contributions

- A QP formulation for real-time squeeze-force allocation between two cooperating arms with friction-aware bounds.
- A motion-adaptive minimal-force term ("task-on-demand") that couples the desired grasp-force floor to the object's motion state, contrasted with a constant-safety-margin baseline.
- A joint-torque-capacity-aware extension of the same QP that biases allocation toward the arm with more actuation margin, added as a quadratic term with no increase in problem size or solver cost.
- Identification and a first mitigation (activation-blended gain switching) for a coupling failure mode between

impedance stiffness and wrench gain during lateral translation.
- Validation on a dual xArm7 platform (mc_rtc).

RELATED WORK

Internal/external force decomposition and QP-based load distribution

- Classical hybrid two-arm coordination and internal/external wrench split (Uchiyama & Dauchez).
- QP-based minimization of internal forces in cooperating manipulators (Nahon & Angeles, 1992).
- NLP/QP force-distribution schemes with normal-force constraints (Kwon & Lee; Kazerooni).
- Recent result: non squeezing internal load distributions from a generalized inverse of the grasp matrix are **not unique**
- Admittance/QP dual-arm cobot frameworks regulating internal vs. external effort for safe bimanual interaction (BAZAR-style hierarchical QP).

Friction-cone-constrained grasp/contact force optimization

- Canonical exact formulation: friction-cone feasibility as positive-definiteness / SDP (Buss, Hashimoto & Moore, 1996).
- Polyhedral/linear approximations of the friction cone for LP/QP tractability (Kerr & Roth; Cheng & Orin).

Motion and slip-adaptive grip force control

- Biological grounding: grip force/load force coordination and friction-dependent safety margin (Westling & Johansson, 1984).
- Trajectory modulation as an alternative (not just complement) to grip-force modulation for slip prevention (Nature Machine Intelligence, 2025)
  contrast: this paper treats grip force as the adaptive variable and trajectory as given, the inverse framing.
- Human hand-acceleration modulation alongside grip-force modulation for slip prevention (2024 study)
  motivates a velocity/acceleration-estimated demand term.
- Reactive slip control via internal-force optimization for multifingered hands, arguing uniform force increases perturb object pose vs. allocation-aware optimization (Théo Ayral)

SYSTEM OVERVIEW

Platform and FSM

- Dual xArm7 setup in mc_rtc.
- FSM: Idle $\rightarrow$ Independent (reach) $\rightarrow$ Collaborative Build-Grasp $\rightarrow$ Trajectory $\rightarrow$ Cooperative Motion .
- One system-diagram figure.

Problem Statement

- At every control cycle, find the per-arm normal (squeeze) forces

$$\mathbf{x} = \begin{bmatrix} F_{n,L} \\ F_{n,R} \end{bmatrix} \tag{1}$$

that:
  - track a desired internal grasp force level,
  - respect friction-derived and actuation bounds,
  - remain balanced between the two arms,

INTERNAL WRENCH DECOMPOSITION AND SQUEEZE ISOLATION

System Wrench and Grasp Matrix

- dual-arm wrench $w \in \mathbb{R}^{1\times 12}$:

$$w = \begin{bmatrix} w_L \\ w_R \end{bmatrix} \in \mathbb{R}^{12}, \tag{2}$$

  where $w_L = [\tau_L; f_L]$ and $w_R = [\tau_R; f_R]$.
- Grasp matrix $G \in \mathbb{R}^{6\times 12}$ and internal-wrench nullspace projector:

$$P_{\text{int}} = I_{12} - G^{\dagger} G. \tag{3}$$

- Internal wrench:

$$w_{\text{int}} = P_{\text{int}}\, w. \tag{4}$$

Squeeze Subspace Isolation

- defined the squeeze direction n_s
- Let N denote the orthonormal nullspace basis of G, obtained via singular value decomposition (SVD).
- Normalized, kinematically valid squeeze direction:

$$n_{\text{squeeze}} = \frac{NN^T n_s}{\|NN^T n_s\|}. \tag{5}$$

Scalar Grasp Force

- The grasp-force scalar is given by

$$\begin{aligned} \lambda_{\text{grasp}} &= n_{\text{squeeze}}^T w_{\text{int}} && (6) \\ &= n_{\text{squeeze}}^T P_{\text{int}}\, w. && (7) \end{aligned}$$

- Since $n_{\text{squeeze}} \in \mathcal{N}(G)$, it follows that $P_{\text{int}} n_{\text{squeeze}} = n_{\text{squeeze}}$. Moreover, P_{int} is an orthogonal projector and is therefore symmetric, i.e., $P_{\text{int}}^T = P_{\text{int}}$. Hence,

$$\begin{aligned} \lambda_{\text{grasp}} &= n_{\text{squeeze}}^T P_{\text{int}}\, w && (8) \\ &= (P_{\text{int}} n_{\text{squeeze}})^T\, w && (9) \\ &= n_{\text{squeeze}}^T w. && (10) \end{aligned}$$

- This identity shows that only the component of w lying in the internal-wrench subspace contributes to λ_{grasp}; any object-motion wrench is orthogonal to n_{squeeze}.

Decomposing w into Fixed and Optimized Parts

- The total contact wrench is decomposed as

$$w = w_{\text{fixed}} + S_n x, \tag{11}$$

  where w_{fixed} contains the measured contact moments and tangential forces (which are not optimized), and $S_n \in \mathbb{R}^{12\times 2}$ maps the decision variables to the world-frame normal-force wrench components through the unit contact normals $\hat{u}_L$ and $\hat{u}_R$.

- Substituting the above decomposition into $\lambda_{\text{grasp}} = n_{\text{squeeze}}^T w$ yields the affine grasp-force model:

$$\lambda_{\text{grasp}}(x) = c^T x + \lambda_0, \tag{12}$$

where

$$c^T = n_{\text{squeeze}}^T S_n, \tag{13}$$

$$\lambda_0 = n_{\text{squeeze}}^T w_{\text{fixed}}. \tag{14}$$

- The coefficients c_L and c_R quantify the contribution of a unit normal force applied by the left and right manipulators, respectively, to the internal squeeze force. The offset λ_0 represents the squeeze already generated by the fixed contact moments and tangential force components.

Motion-Adaptive Force Demand

Motivation

- Constant-safety-margin baseline is simple but not motion-aware: over-conservative at rest, potentially insufficient at peak dynamic load.
- Design goal: the minimum grasp-force floor should increase during object transport and relax when the object is at rest.

Trajectory-Schedule-Driven Formulation

- Considered index: acceleration profile Rejected because $|\ddot{s}(\tau)|$ exhibits a double-lobe profile that returns to zero at peak velocity, despite cumulative inertial and frictional loading still being present.
- Adopted index: velocity profile This profile is single-lobed, continuous, and zero only at the trajectory endpoints. The choice is deliberate and should be justified explicitly in the text.
- measured or estimated task-space velocity signal, e.g., the average of the commanded or measured end-effector velocities:

$$v_{\text{obj,est}} = \left\| \frac{1}{2} \left(\dot{x}_L + \dot{x}_R \right) \right\| \tag{15}$$

$$F_{\text{demand}} = k \, v_{\text{obj,est}} \tag{16}$$

- Since the formulation relies on a measured signal rather than a planner-specific trajectory clock, it generalizes across different trajectory generators
- **Optional for future works** hybrid formulation: use the schedule-driven term as a smooth feedforward baseline with a small proportional correction based on the planned-versus-measured velocity error. This configuration is proposed as the ablation condition (c) in Section X.

Minimal Grasp Force Bound

- Minimum normal-force bound (per arm):

$$F_{\min} = \max\left(\frac{\|F_t\|}{\mu}, \; F_{\text{static}} + F_{\text{demand}} \right). \tag{17}$$

- Interpretation: the minimum normal-force floor is given by the larger of the friction-required minimum and the combined static and motion-adaptive safety margin.

QP Formulation for Squeeze Force Optimization

Cost Function

- Objective function:

$$J(x) = \underbrace{\alpha \, \lambda_{\text{grasp}}(x)^2}_{\text{minimize squeeze}} + \underbrace{\beta \left(F_{n,L} - F_{n,R} \right)^2}_{\text{balance}}, \qquad \alpha, \beta > 0. \tag{18}$$

- α: penalizes the total squeeze effort (crushing risk, actuation cost).
- β: penalizes load imbalance between the two arms (avoids one arm doing all the work).

Standard QP Form Derivation

- Expanding the grasp-force term:

$$\lambda_{\text{grasp}}(x)^2 = x^T c c^T x + 2\lambda_0 c^T x + \lambda_0^2. \tag{19}$$

- Defining $d = [1, -1]^T$, the balance term is written as

$$\left(F_{n,L} - F_{n,R} \right)^2 = x^T d d^T x. \tag{20}$$

- Dropping the constant term $\alpha \lambda_0^2$, the objective becomes

$$J(x) = x^T \left(\alpha c c^T + \beta d d^T \right) x + 2\alpha\lambda_0 c^T x. \tag{21}$$

- Matching the standard QP form

$$J(x) = \frac{1}{2} x^T H x + g^T x, \tag{22}$$

yields

$$H = 2 \begin{bmatrix} \alpha c_L^2 + \beta & \alpha c_L c_R - \beta \\ \alpha c_L c_R - \beta & \alpha c_R^2 + \beta \end{bmatrix}, \tag{23}$$

and

$$g = 2\alpha\lambda_0 \begin{bmatrix} c_L \\ c_R \end{bmatrix}. \tag{24}$$

Constraints

- Friction constraints (per contact):

$$\|F_{t,L}\| \le \mu_L F_{n,L}, \tag{25}$$

$$\|F_{t,R}\| \le \mu_R F_{n,R}, \tag{26}$$

where the tangential contact forces are obtained from the force sensors and are treated as fixed (i.e., not optimization variables).
- Positivity / minimum-force constraints:

$$F_{n,L} \ge F_{\min}, \tag{27}$$

$$F_{n,R} \ge F_{\min}, \tag{28}$$

- The friction constraints use the tangential force measured during the *previous* control cycle to constrain the *next* commanded normal force, introducing a one-cycle lag. This is a deliberate linear/QP approximation of the exact friction-cone constraint, avoiding the need for a jointly optimized SOCP formulation (Section II-B).

Final Problem

- The optimization problem is formulated as

$$\min_{x} \ J(x) \tag{29}$$

subject to

$$F_{n,L} \geq F_{\min}, \tag{30}$$
$$F_{n,R} \geq F_{\min}, \tag{31}$$

together with the upper actuation bounds.

Sensitivity to α, β

- Small figure or table: sweep α (with fixed β) and β (with fixed α), and report the resulting λ_{grasp} tracking performance and inter-arm load imbalance. This provides intuition for parameter tuning and serves as a robustness check.

Joint-Torque-Capacity-Aware Extension

Motivation

- The QP formulated thus far allocates squeeze force exclusively at the end-effector wrench level and does not account for the joint-space cost associated with each allocation.
- End-effector wrenches are mapped to joint torques through the Jacobian transpose,

$$\tau = J^T w. \tag{32}$$

Since w is already an affine function of x, the resulting joint torques are also affine in x, allowing this information to be incorporated without introducing additional decision variables.
- The objective is to bias the squeeze-force allocation toward the manipulator with greater available actuation margin (e.g., farther from a singular configuration, subject to lower thermal loading, or with greater torque headroom), while preserving the original real-time QP structure.

Affine Torque Map

- Per-arm contact wrench:

$$w_L = w_{fixed,L} + S_{n,L} F_{n,L},$$
$$w_R = w_{fixed,R} + S_{n,R} F_{n,R}.$$

- Corresponding joint-torque contributions:

$$\tau_{c,L}(x) = J_L^T w_{fixed,L} + J_L^T S_{n,L} F_{n,L},$$
$$\tau_{c,R}(x) = J_R^T w_{fixed,R} + J_R^T S_{n,R} F_{n,R},$$

which are affine in x, with the same structure as $\lambda_{grasp}(x)$ in section number.

Extended Cost Function

$$J(x) = \alpha \, \lambda_{grasp}(x)^2 + \beta \, (F_{n,L} - F_{n,R})^2 \tag{33}$$
$$+ \gamma_L \|\tau_{c,L}(x)\|^2 + \gamma_R \|\tau_{c,R}(x)\|^2 \tag{34}$$

- Each new term is quadratic in x no change to problem dimension (still 2 variables) or solver.
- Term-by-term reading of the four-way trade-off: α (avoid crushing the object), β (keep contact forces even), γ_L, γ_R (avoid joint torque overload on each arm respectively).

Role and Tuning of γ_L, γ_R

- $\gamma_L = \gamma_R = 0$ recovers the original formulation exactly the extension is strictly backward-compatible, useful for a clean ablation (with/without torque awareness) rather than a redesign.
- Small, equal, nonzero values (e.g., 0.01–0.1) protect joint torque margin without materially disrupting grasp-force tracking report the smallest γ that yields a measurable protective effect.
- Asymmetric values ($\gamma_L \neq \gamma_R$) let the allocator favor the arm with more capacity e.g., setting γ_L higher than γ_R to shift squeeze effort onto the right arm when the left arm is known (a priori, per configuration/experiment) to be closer to a singularity, torque limit, or thermal derate.

Target Formulation (future work)

- Reformulate as orthogonal task-space separation: decompose the task space into complementary force-controlled and motion-controlled subspaces so spring/wrench-gain terms and lateral motion-tracking terms act on disjoint directions by construction, rather than being reconciled post hoc via gain blending.
- Cite the classical hybrid position/force orthogonal-subspace formulation (Raibert & Craig, 1981) and a recent interaction-frame instance (2026) as the target formalism.

Experimental Validation

Platform and Protocol

- Dual xArm7, `mc_rtc`.
- Pick–carry–place trajectory; force-demand conditions:
  - (a) constant `F_static` baseline,
  - (b) schedule-driven `F_demand`,
  - (c) motion-estimated `F_demand`,
  - (d) hybrid.
- Torque-awareness conditions: with vs. without the extension, symmetric vs. asymmetric γ_L, γ_R.

Metrics

- Internal force
- Peak/RMS tangential force relative to the friction bound (slip-margin proxy).
- Total squeeze effort $\int \|x\| \, dt$.

- Peak/RMS joint torque per arm, and margin-to-limit, with vs. without extension.
- QP solve time distribution

Figures

- FSM / system diagram.
- Internal force vs. trajectory phase, with all demand conditions overlaid.
- Friction-margin utilization over the trajectory.
- Joint torque per arm, with vs. without torque-aware weighting, under an induced asymmetric-capacity scenario.
- Contact-loss / activation-blend before-after comparison.
- Solve-time histogram.

Discussion and Limitations

- Nonuniqueness of the internal/external decomposition: the weighted objective is a design choice, not "the" optimal split. State what it privileges (low effort vs. balance vs. torque margin) and how that choice was made.
- Linearized, one-cycle-lagged friction bound versus the exact friction cone: identify the operating regime in which this approximation may fail (e.g., fast tangential-force transients).
- Fixed versus state-driven weighting coefficients, γ_L and γ_R: fixed weights are straightforward to justify but do not capture the instantaneous actuation capacity of each manipulator. **in future works** to present state-dependent weighting as the natural next step.

Conclusion and Future Work

- **Summary:** The proposed framework combines QP-based dual-arm squeeze-force allocation, a motion-adaptive force-demand formulation, and a joint-torque-capacity-aware extension, and is validated experimentally on a dual `xArm7` platform.
- **Future Work:**
  - Adopt the closed-loop, motion-estimated force-demand formulation as the primary approach, rather than as a fallback strategy.
  - Replace the fixed weighting coefficients γ_L and γ_R with state-driven weights based on online measures such as manipulability, joint-torque headroom, or actuator thermal state.
  - Incorporate exact friction-cone handling through a second-order cone programming (SOCP) formulation, subject to real-time computational constraints.

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  - Recent analysis on the nonuniqueness of nonsqueezing load distributions in cooperative manipulation (2023).
  - Admittance-QP dual-arm cobot framework regulating internal/external efforts (BAZAR, 2017–2018).
  - Recent HQP variants for mobile/free-floating dual-arm manipulation (2025–2026).
- **Grasp/contact force optimization under friction**
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  - Kerr, J., and Roth, B.; Cheng, F., and Orin, D. — polyhedral/linear friction-cone approximations.
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  - Human hand-acceleration modulation study for slip prevention (2024).
  - Reactive slip control via internal-force optimization for multifingered hands (2026 arXiv).
- **Joint-space capacity-aware allocation**
  - Classical minimum-norm / weighted-pseudoinverse torque allocation for redundant and multi-arm systems.
  - Whole-body and multi-contact QP controllers with torque- or actuation-margin-aware cost terms (locomotion / multi-contact force-distribution literature).
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